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Course code:CSA0604

Course name: Data Handling and Visualization

Set 21: Energy Consumption Data

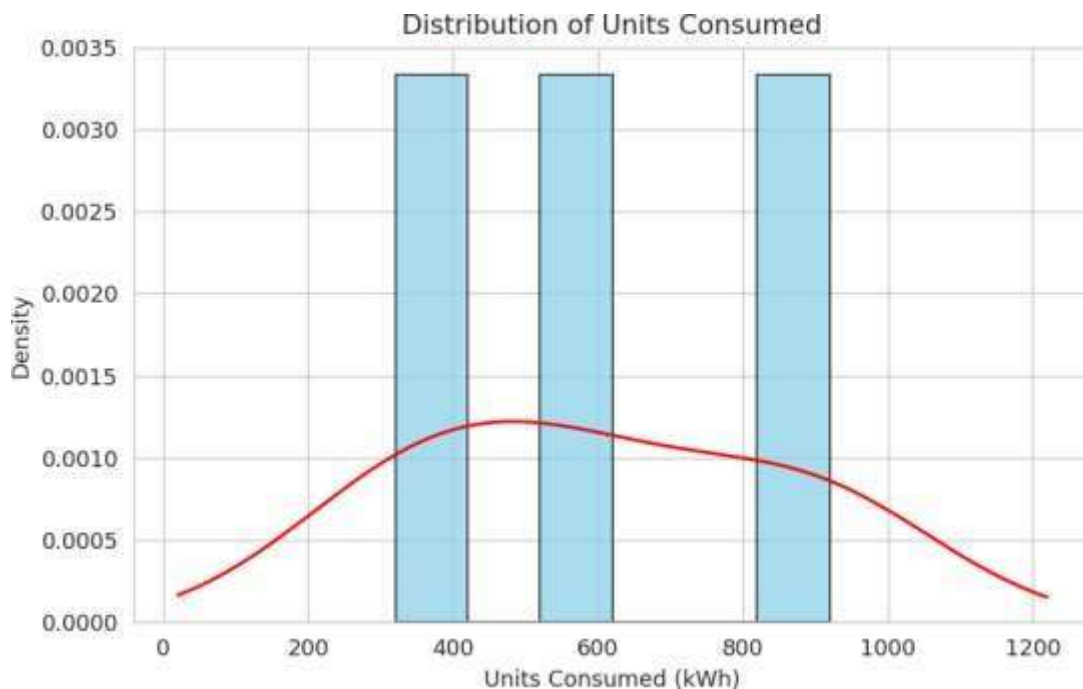
Question 1

Import the dataset in R. Plot a histogram and density plot for `Units_Consumed`. Compare consumption patterns.

R Code

```
# 1. Histogram and density plot for Units_Consumed
ggplot(df, aes(x = Units_Consumed)) +
  geom_histogram(aes(y = after_stat(density)), binwidth = 100, fill = "skyblue", color = "black") +
  geom_density(color = "red", linewidth = 1) +
  labs(title = "Distribution of Units Consumed", x = "Units Consumed (kWh)", y = "Density") +
  theme_minimal()
ggsave("histogram_density_units_consumed.png", width = 7, height = 5)
cat("Comparison: Density plot shows the spread and skew of energy consumption across records.\n")
```

Output



Question 2

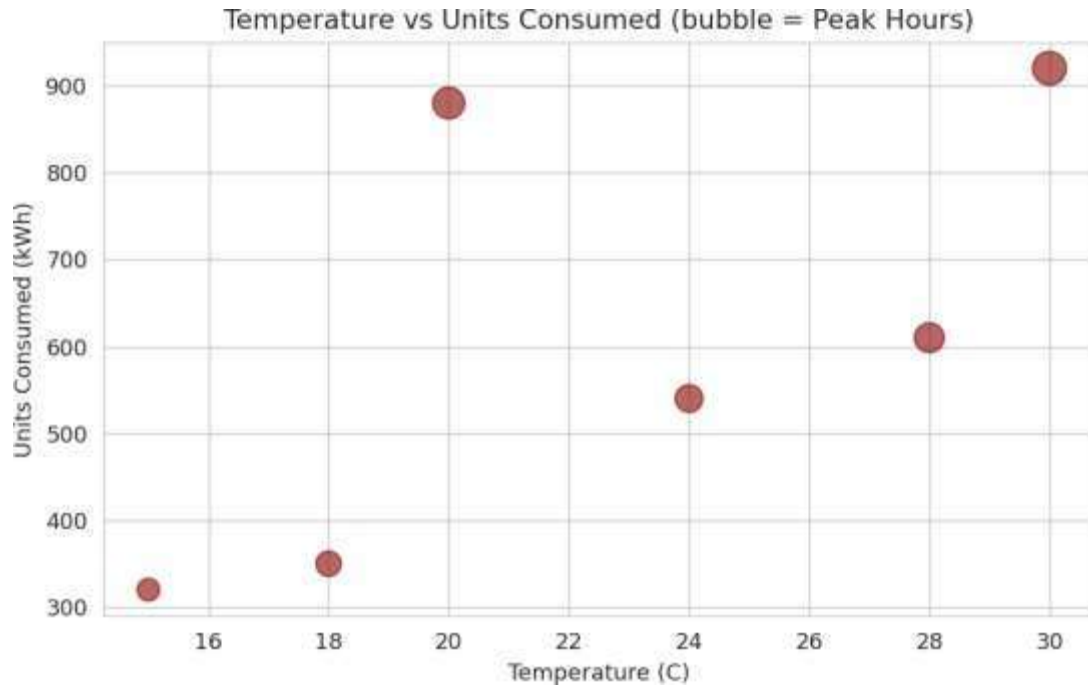
Using R programming, create a scatter plot between `Temperature` and `Units_Consumed`. Represent `Peak_Hours` using bubble size. Apply transparency.

R Code

```
# 2. Scatter plot - Temperature vs Units_Consumed, bubble size = Peak_Hours
ggplot(df, aes(x = Temperature, y = Units_Consumed, size = Peak_Hours)) +
  geom_point(alpha = 0.6, color = "darkred") +
```

```
labs(title = "Temperature vs Units Consumed (Bubble = Peak Hours)",
x = "Temperature (C)", y = "Units Consumed (kWh)") +
theme_minimal()
ggsave("scatter_temp_vs_units.png", width = 7, height = 5)
```

Output



Question 3

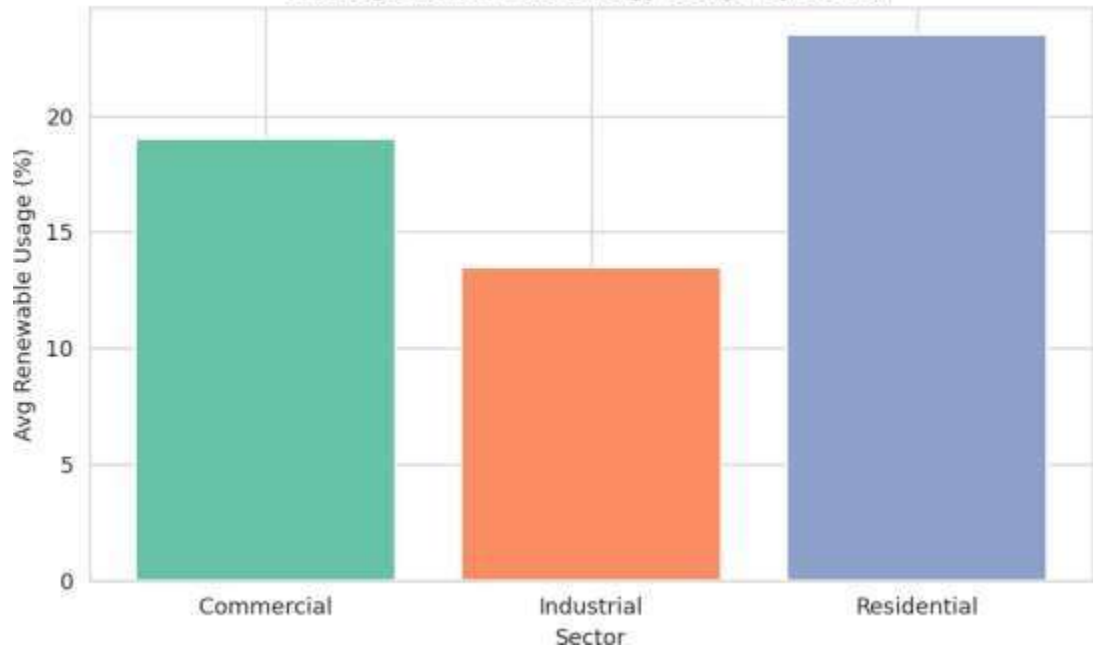
Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

R Code

```
# 3. Average Renewable_Usage per Sector - bar chart
sector_avg <- df %>% group_by(Sector) %>% summarise(AvgRenewable = mean(Renewable_Usage))
ggplot(sector_avg, aes(x = Sector, y = AvgRenewable, fill = Sector)) +
geom_bar(stat = "identity") +
labs(title = "Average Renewable Energy Usage by Sector", x = "Sector", y = "Avg Renewable Usage (%)") +
theme_minimal()
ggsave("bar_chart_renewable_usage.png", width = 7, height = 5)
cat("Interpretation: Sectors with higher average renewable usage are adopting greener energy sources more.\n")
```

Output

Average Renewable Energy Usage by Sector



Set 22: Time Series Analysis

Question 1

In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

R Code

```
# 1. Time series line chart - trend in monthly sales
sales_ts <- ts(df$Sales, start = 1, frequency = 1)
png("timeseries_line_chart_sales.png", width = 700, height = 500)
plot(sales_ts, type = "o", col = "blue", main = "Monthly Sales Trend",
     xlab = "Month", ylab = "Sales ($)", xaxt = "n")
axis(1, at = 1:5, labels = df$Month)
dev.off()
```

Output



Question 2

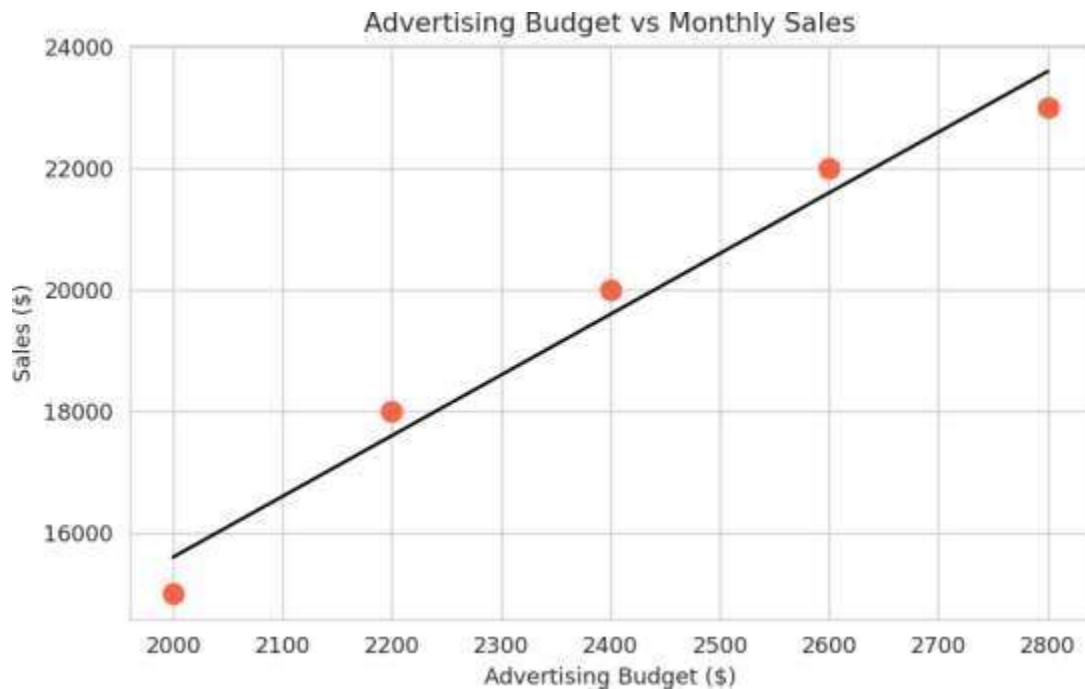
In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

R Code

```
# 2. Scatter plot - advertising budget vs monthly sales
ggplot(df, aes(x = AdvertisingBudget, y = Sales)) +
  geom_point(color = "tomato", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "Advertising Budget vs Monthly Sales", x = "Advertising Budget ($)", y = "Sales ($)") +
  theme_minimal()
```

```
ggsave("scatter_ad_budget_vs_sales.png", width = 7, height = 5)
cat("Insight: Correlation =", cor(df$AdvertisingBudget, df$Sales), "\n")
```

Output



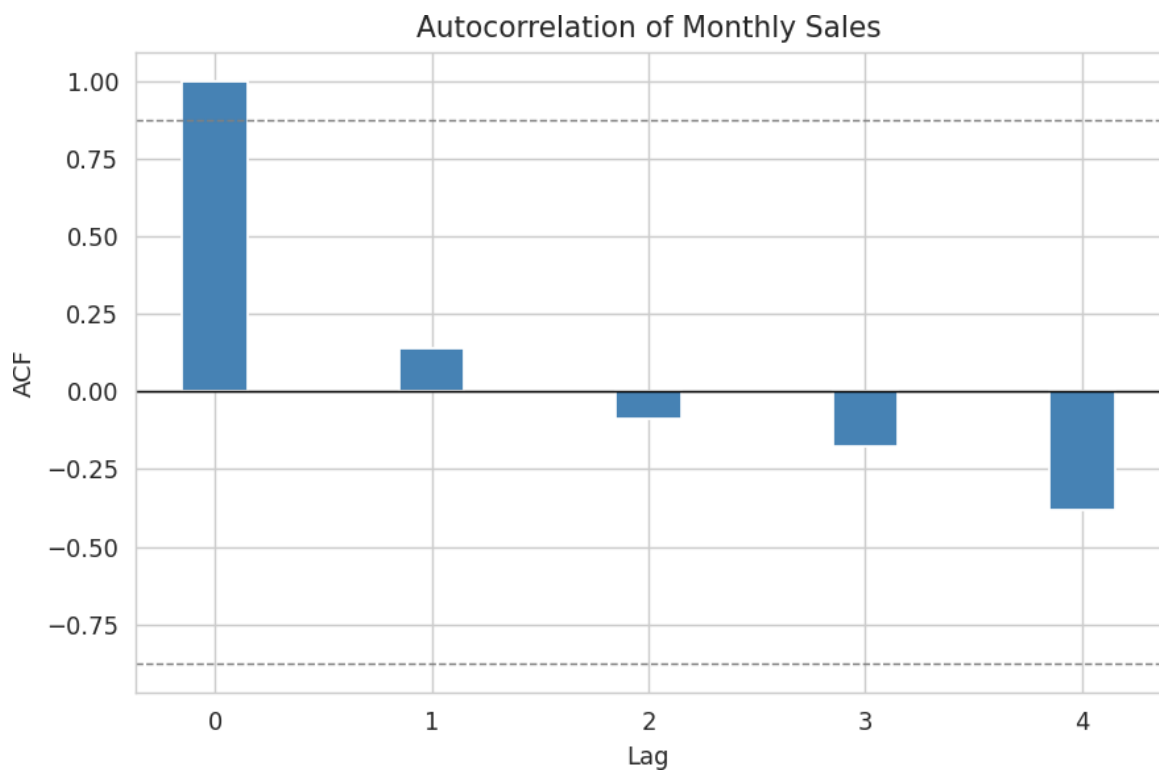
Question 4

In R, create an autocorrelation plot to identify seasonality in the time series data.

R Code

```
# 4. Autocorrelation plot to identify seasonality
png("acf_plot_sales.png", width = 700, height = 500)
acf(df$Sales, main = "Autocorrelation of Monthly Sales")
dev.off()
```

Output



Set 23: Employee Performance Analysis

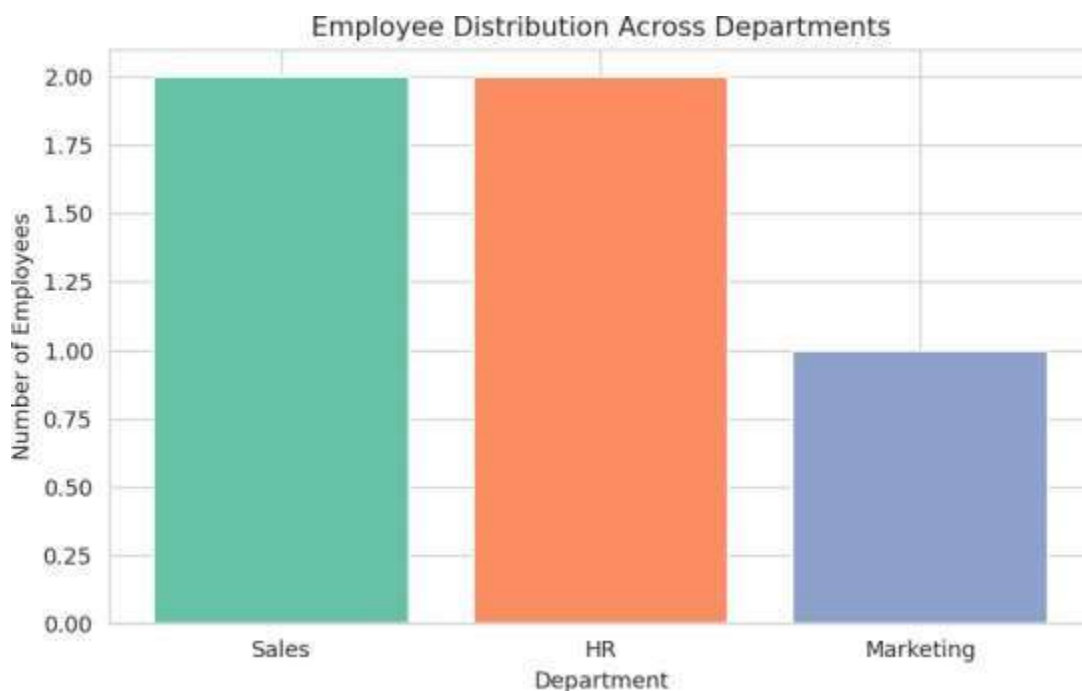
Question 1

In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

R Code

```
# 1. Bar chart - distribution of employees across departments (R)
dept_counts <- df %>% distinct(EmployeeID, Department) %>% count(Department)
ggplot(dept_counts, aes(x = Department, y = n, fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Employee Distribution by Department", x = "Department", y = "Count") +
  theme_minimal()
ggsave("bar_chart_department_distribution.png", width = 7, height = 5)
```

Output



Question 2

In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

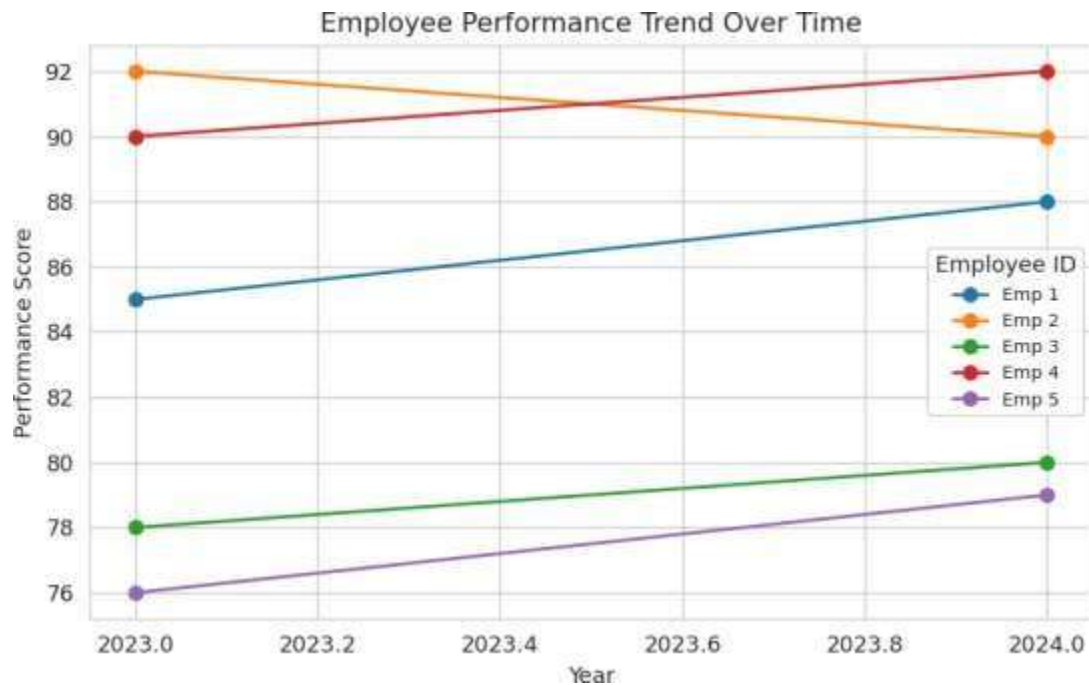
R Code

```
# 2. Line chart - performance trend of employees over time (R)
ggplot(df, aes(x = Year, y = PerformanceScore, color = factor(EmployeeID), group = EmployeeID)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(title = "Employee Performance Trend", x = "Year", y = "Performance Score", color = "Employee ID") +
```



```
theme_minimal()
ggsave("line_chart_performance_trend.png", width = 7, height = 5)
```

Output



Question 4

In R, develop a table showing the performance data for each employee. Label the table elements.

R Code

```
# 4. Table - performance data for each employee (R)
perf_table <- df %>% gt() %>% tab_header(title = "Employee Performance Data")
gtsave(perf_table, "table_employee_performance.png")
print(df)
```

Output

EmployeeID	Department	YearsOfService	PerformanceScore	Year
1	Sales	5	85	2023
2	HR	3	92	2023
3	Marketing	7	78	2023
4	Sales	4	90	2023
5	HR	2	76	2023

Set 24: Product Inventory Management

Question 1

In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

R Code

```
# 1. Bar chart - quantity of each product in inventory (R)
ggplot(df, aes(x = ProductName, y = QuantityAvailable, fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(title = "Quantity of Each Product in Inventory", x = "Product", y = "Quantity Available") +
  theme_minimal()
ggsave("bar_chart_inventory_quantity.png", width = 7, height = 5)
```

Output



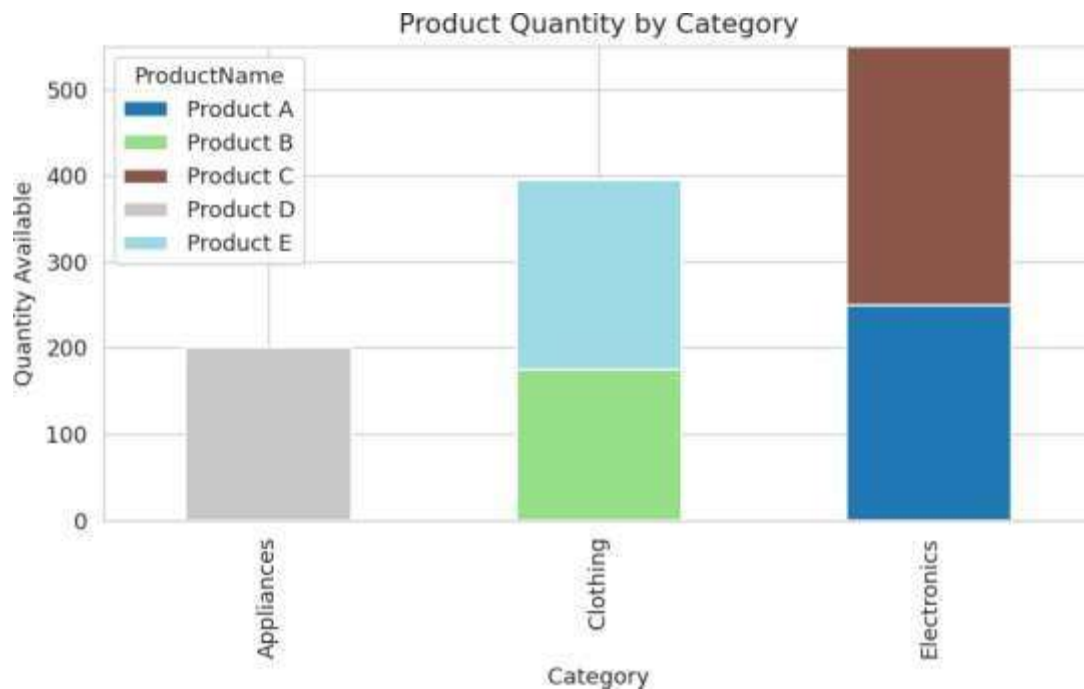
Question 2

In R, generate a stacked bar chart to show the quantity of each product within different product categories.

R Code

```
# 2. Stacked bar chart - quantity within different categories (R)
ggplot(df, aes(x = Category, y = QuantityAvailable, fill = ProductName)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(title = "Product Quantity by Category", x = "Category", y = "Quantity Available") +
  theme_minimal()
ggsave("stacked_bar_quantity_by_category.png", width = 7, height = 5)
```

Output



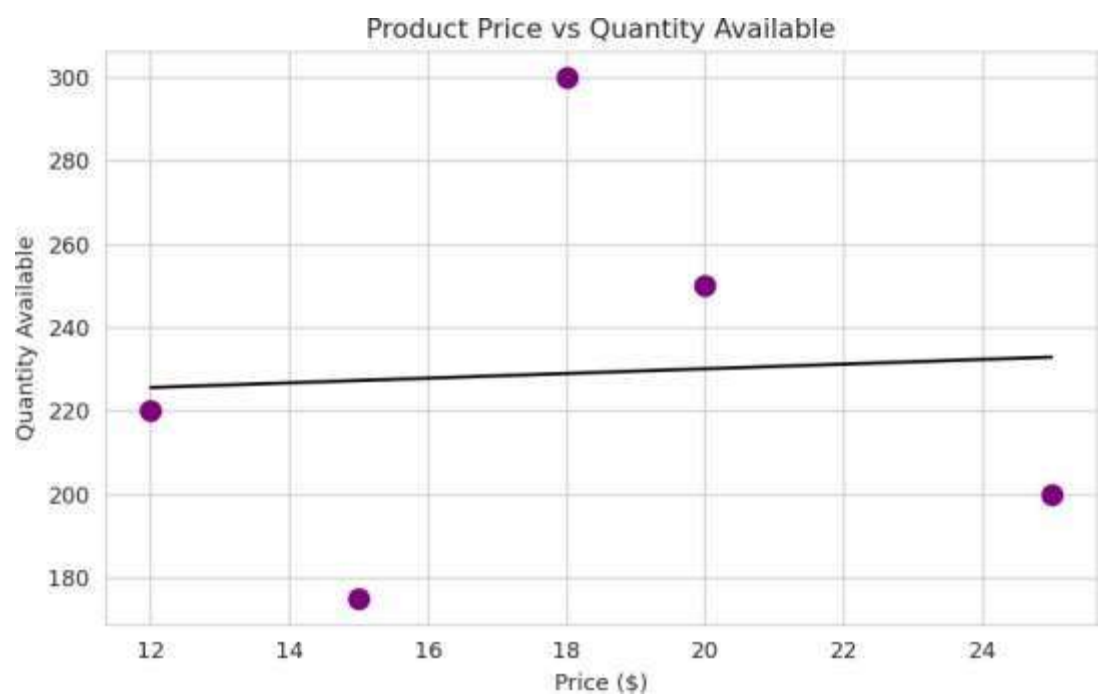
Question 4

In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

R Code

```
# 4. Scatter plot - price vs quantity available (R)
ggplot(df, aes(x = Price, y = QuantityAvailable)) +
  geom_point(color = "purple", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "Product Price vs Quantity Available", x = "Price ($)", y = "Quantity Available") +
  theme_minimal()
ggsave("scatter_price_vs_quantity.png", width = 7, height = 5)
cat("Insight: Correlation =", cor(df$Price, df$QuantityAvailable), "\n")
```

Output



Set 25: Website Traffic Analysis

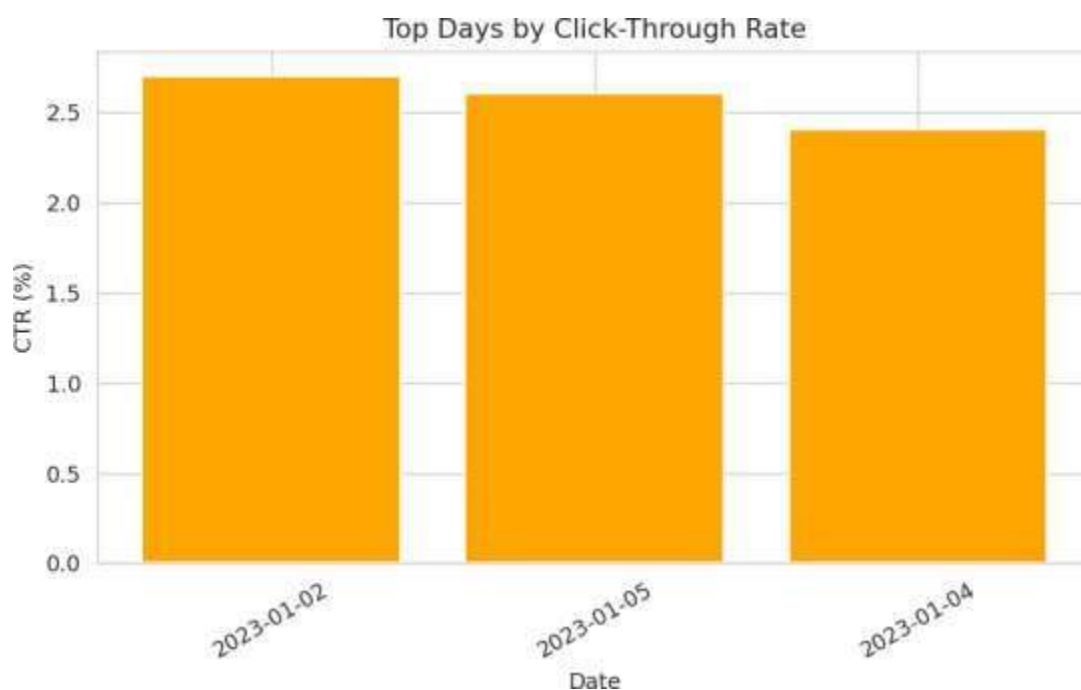
Question 2

In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

R Code

```
# 2. Bar chart (R) - top N days with highest click-through rates
top_n_days <- df %>% arrange(desc(ClickThroughRate)) %>% head(3)
ggplot(top_n_days, aes(x = reorder(as.character(Date), -ClickThroughRate), y = ClickThroughRate)) +
  geom_bar(stat = "identity", fill = "orange") +
  labs(title = "Top Days by Click-Through Rate", x = "Date", y = "CTR (%)") +
  theme_minimal()
ggsave("bar_chart_top_ctr_days.png", width = 7, height = 5)
```

Output



Question 3

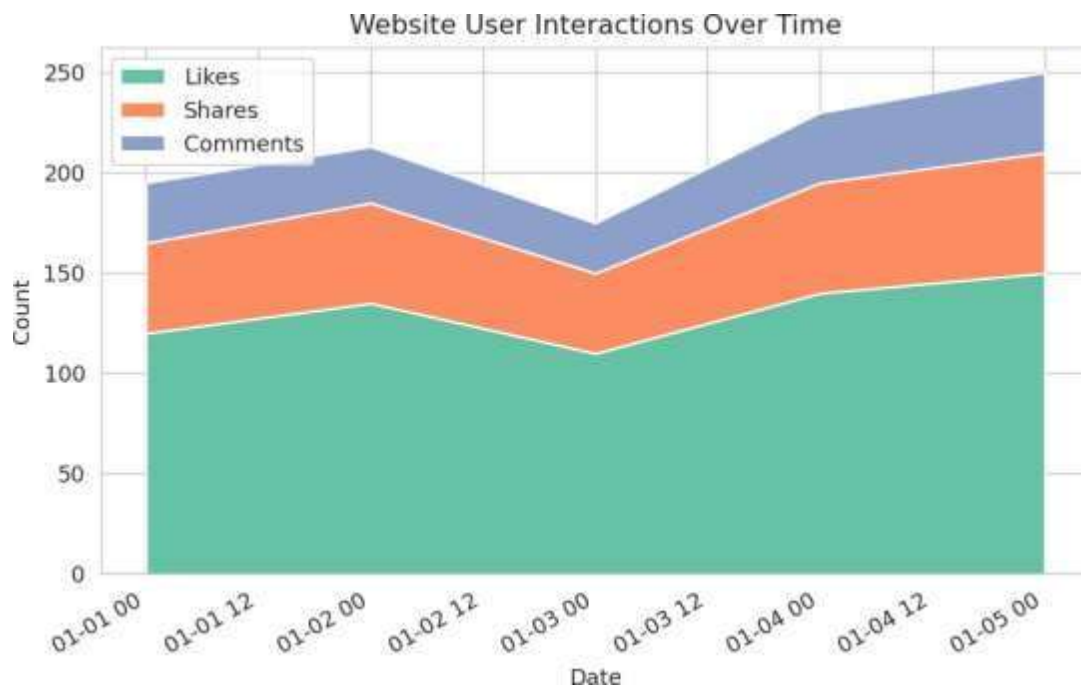
In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

R Code

```
# 3. Stacked area chart (R) - user interactions (likes, shares, comments)
df_long <- df %>% pivot_longer(cols = c(Likes, Shares, Comments), names_to = "Interaction",
  values_to = "Count")
ggplot(df_long, aes(x = Date, y = Count, fill = Interaction)) +
  geom_area(position = "stack") +
  labs(title = "Website User Interactions Over Time", x = "Date", y = "Count") +
```

```
theme_minimal()
ggsave("stacked_area_interactions.png", width = 7, height = 5)
```

Output



Set 26: Student Mini Data

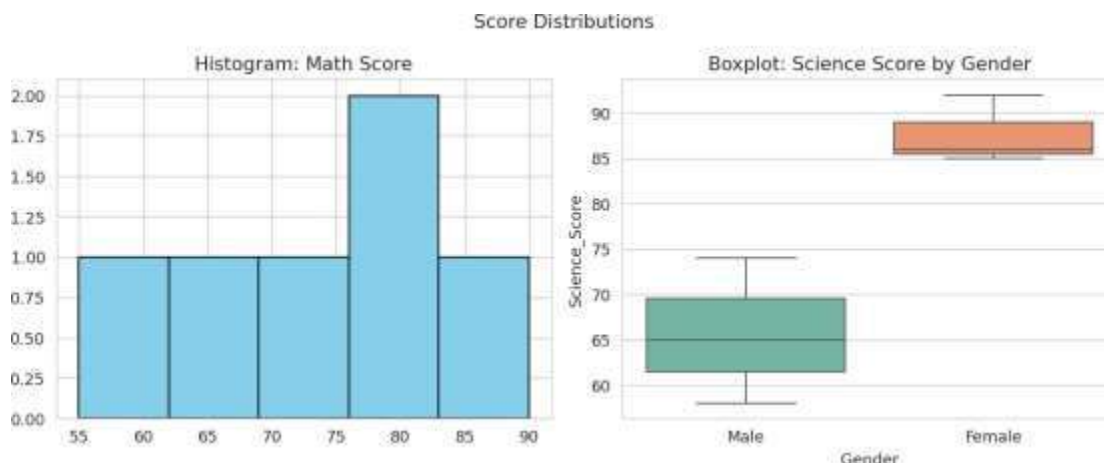
Question 1

Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.

R Code

```
# 1. Histogram for Math_Score + boxplot of Science_Score by Gender
ggplot(df, aes(x = Math_Score)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Math Scores", x = "Math Score", y = "Frequency") +
  theme_minimal()
ggsave("histogram_math_score.png", width = 7, height = 5)
ggplot(df, aes(x = Gender, y = Science_Score, fill = Gender)) +
  geom_boxplot() +
  labs(title = "Science Score by Gender", x = "Gender", y = "Science Score") +
  theme_minimal()
ggsave("boxplot_science_by_gender.png", width = 7, height = 5)
cat("Interpretation: Compare median and spread of Science_Score between genders.\n")
```

Output



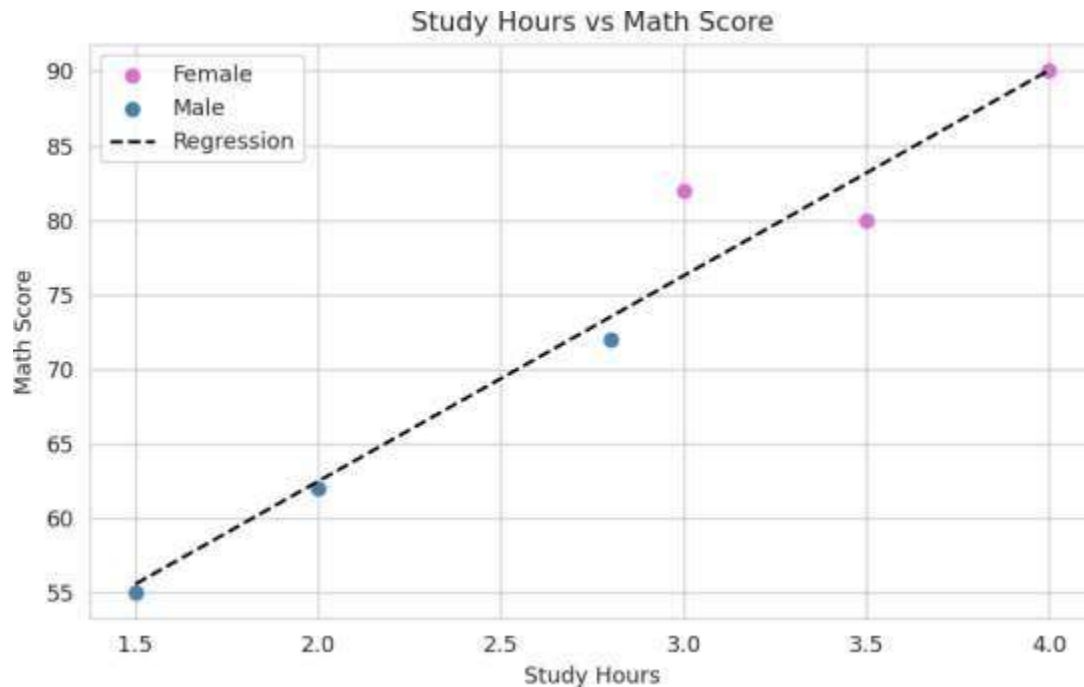
Question 2

Create a scatter plot using R programming between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.

R Code

```
# 2. Scatter plot - Study_Hours vs Math_Score, colored by Gender, with regression line
ggplot(df, aes(x = Study_Hours, y = Math_Score, color = Gender)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Study Hours vs Math Score", x = "Study Hours", y = "Math Score") +
  theme_minimal()
ggsave("scatter_study_hours_vs_math.png", width = 7, height = 5)
cat("Relationship: More study hours are generally associated with higher math scores.\n")
```

Output



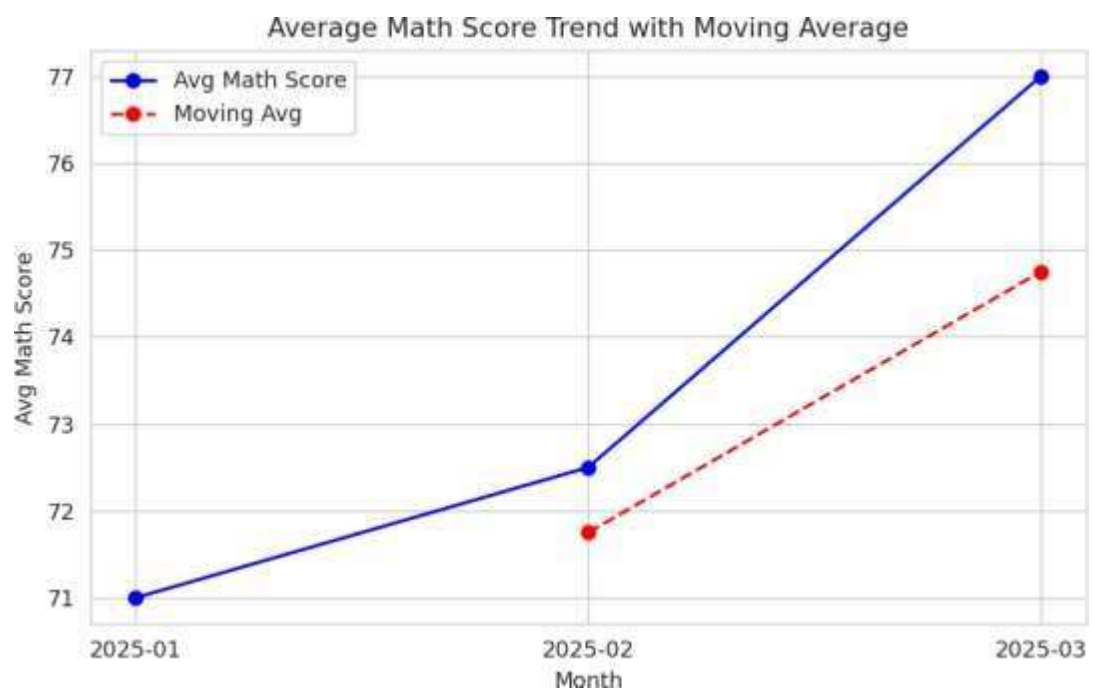
Question 3

Using R, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.

R Code

```
# 3. Convert Exam_Date, monthly average Math scores, line chart with moving average
df$Exam_Date <- as.Date(df$Exam_Date)
df$MonthYear <- format(df$Exam_Date, "%Y-%m")
monthly_avg <- df %>% group_by(MonthYear) %>% summarise(AvgMathScore = mean(Math_Score)) %>%
  arrange(MonthYear)
monthly_avg$MovingAvg <- rollmean(monthly_avg$AvgMathScore, k = 2, fill = NA, align = "right")
ggplot(monthly_avg, aes(x = MonthYear, group = 1)) +
  geom_line(aes(y = AvgMathScore), color = "blue") +
  geom_line(aes(y = MovingAvg), color = "red", linetype = "dashed") +
  labs(title = "Average Math Score Trend with Moving Average", x = "Month", y = "Avg Math Score") +
  theme_minimal()
ggsave("line_chart_avg_math_score_trend.png", width = 7, height = 5)
```

Output



Set 27: Patient Health Risk Analysis

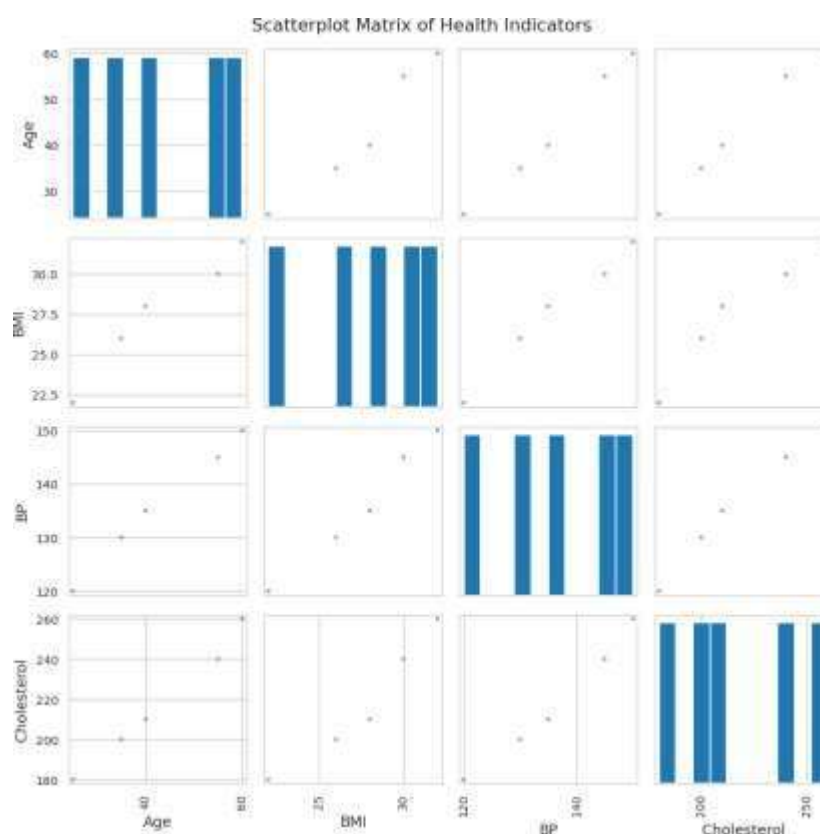
Question 1

Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

R Code

```
# 1. Scatterplot matrix for Age, BMI, BP, Cholesterol
png("scatterplot_matrix_health.png", width = 800, height = 800)
ggpairs(df, columns = c("Age", "BMI", "BP", "Cholesterol"),
  title = "Scatterplot Matrix of Health Indicators")
dev.off()
cat("Interpretation: Look for strong pairwise correlations (e.g., Age vs BP, BMI vs Cholesterol)
that may indicate risk clusters.\n")
```

Output



Question 2

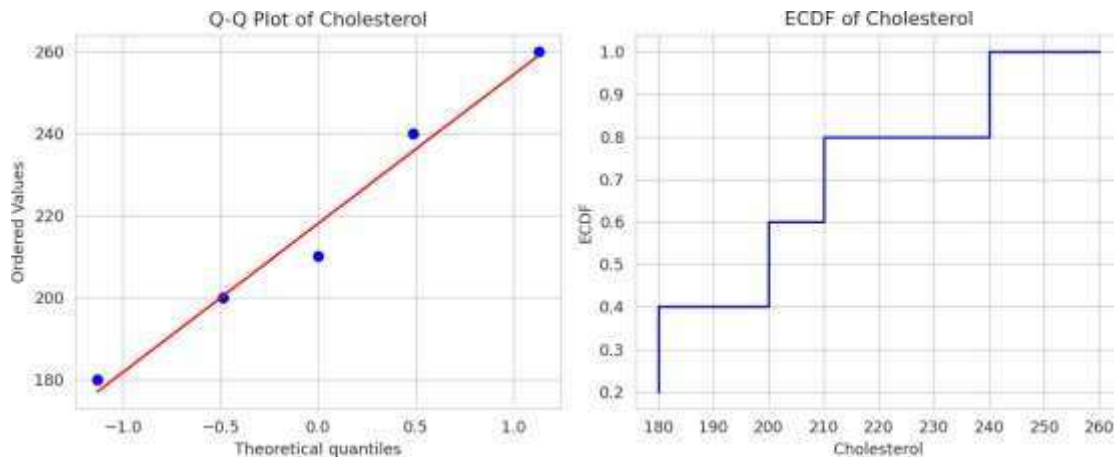
Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

R Code

```
# 2. Q-Q plot and ECDF for Cholesterol
png("qqplot_cholesterol.png", width = 700, height = 500)
```

```
qqnorm(df$Cholesterol, main = "Q-Q Plot of Cholesterol")
qqline(df$Cholesterol, col = "red")
dev.off()
ggplot(df, aes(x = Cholesterol)) +
  stat_ecdf(geom = "step", color = "blue") +
  labs(title = "ECDF of Cholesterol Levels", x = "Cholesterol", y = "ECDF") +
  theme_minimal()
ggsave("ecdf_cholesterol.png", width = 7, height = 5)
cat("Comment: Assess whether points on the Q-Q plot follow the reference line (normality) and
note skewness direction.\n")
```

Output



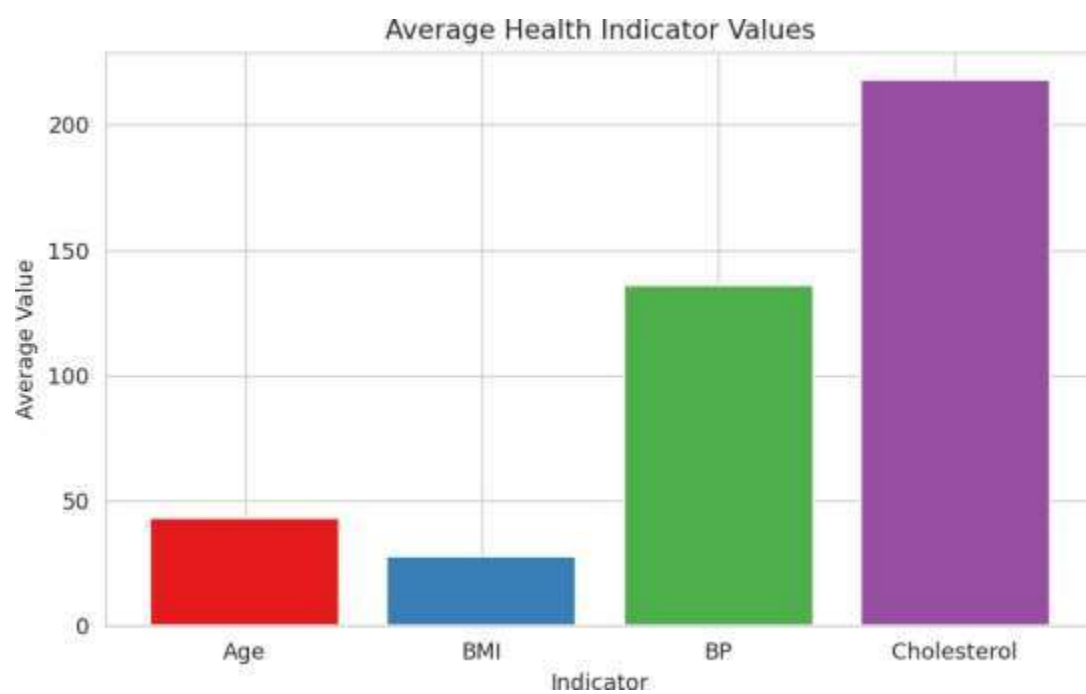
Question 3

Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk.

R Code

```
# 3. Bar chart - average values of each health indicator
avg_values <- data.frame(
  Indicator = c("Age", "BMI", "BP", "Cholesterol"),
  Average = c(mean(df$Age), mean(df$BMI), mean(df$BP), mean(df$Cholesterol))
)
ggplot(avg_values, aes(x = Indicator, y = Average, fill = Indicator)) +
  geom_bar(stat = "identity") +
  labs(title = "Average Health Indicator Values", x = "Indicator", y = "Average Value") +
  theme_minimal()
ggsave("bar_chart_avg_health_indicators.png", width = 7, height = 5)
cat("Interpretation: Higher average BP and Cholesterol relative to normal ranges may indicate
elevated general patient risk.\n")
```

Output



Set 28: Vehicle Performance Analysis

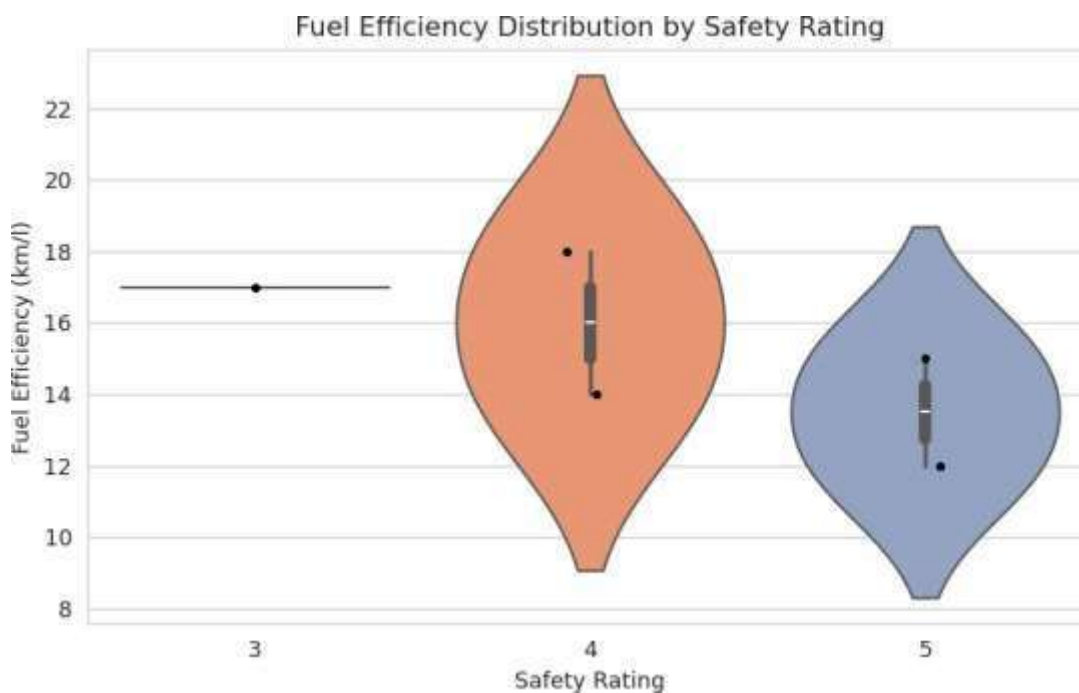
Question 1

Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

R Code

```
# 1. Violin plot - Fuel_Efficiency distribution across Safety_Rating categories
ggplot(df, aes(x = factor(Safety_Rating), y = Fuel_Efficiency, fill = factor(Safety_Rating))) +
  geom_violin() +
  geom_jitter(width = 0.05) +
  labs(title = "Fuel Efficiency Distribution by Safety Rating", x = "Safety Rating", y = "Fuel
Efficiency (km/l)") +
  theme_minimal()
ggsave("violin_fuel_efficiency_by_safety.png", width = 7, height = 5)
cat("Interpretation: Compare spread and median fuel efficiency within each safety rating
group.\n")
```

Output



Question 2

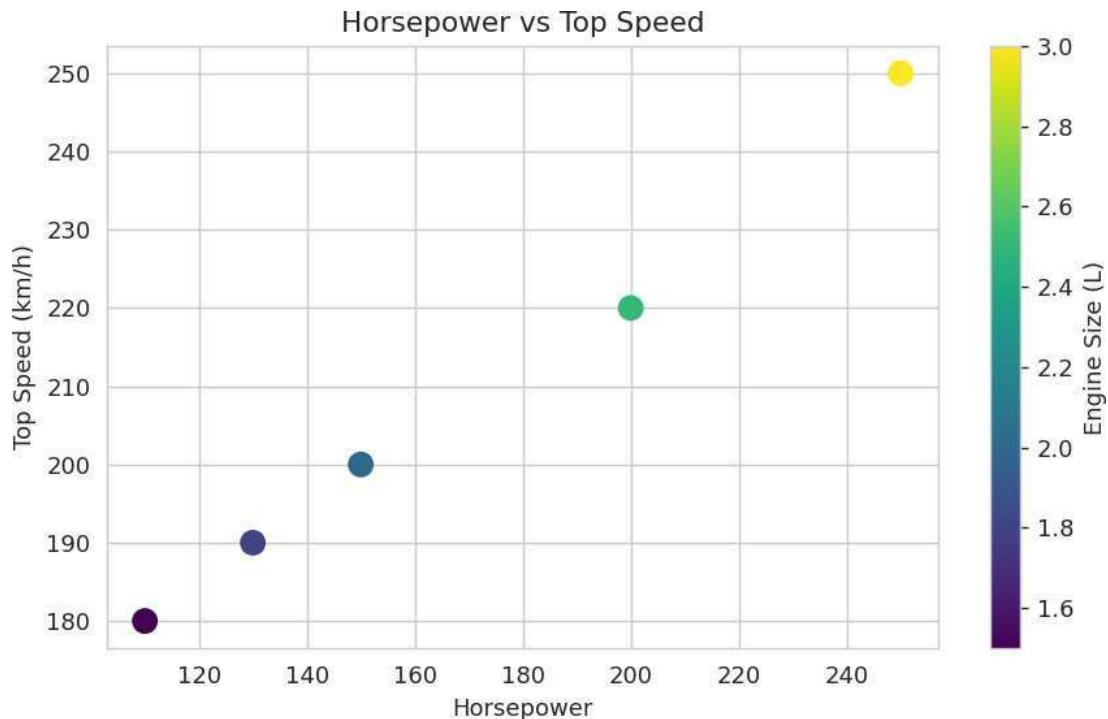
Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

R Code

```
# 2. Scatter plot - Horsepower vs Top_Speed, colored by Engine_Size
```

```
ggplot(df, aes(x = Horsepower, y = Top_Speed, color = Engine_Size)) +
  geom_point(size = 4) +
  labs(title = "Horsepower vs Top Speed", x = "Horsepower", y = "Top Speed (km/h)") +
  theme_minimal()
ggsave("scatter_horsepower_vs_topspeed.png", width = 7, height = 5)
cat("Insight: Correlation between Horsepower and Top Speed =", cor(df$Horsepower, df$Top_Speed),
    "\n")
```

Output



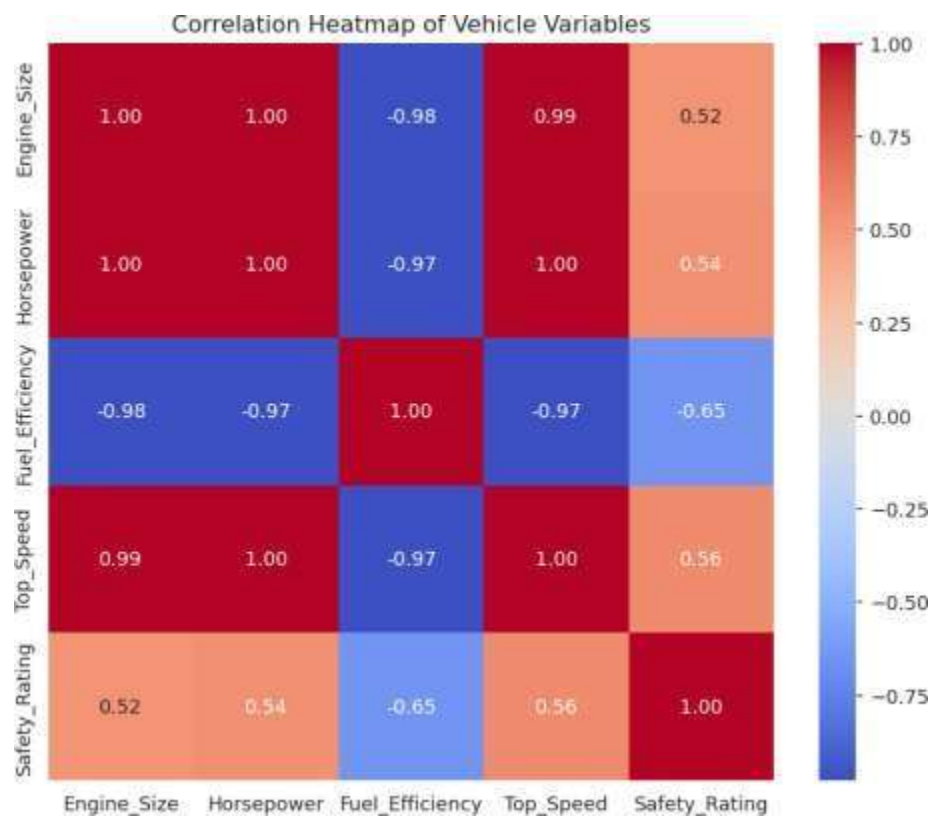
Question 3

Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

R Code

```
# 3. Correlation heatmap for all numerical variables
num_df <- df %>% select(Engine_Size, Horsepower, Fuel_Efficiency, Top_Speed, Safety_Rating)
corr_matrix <- cor(num_df)
corr_melt <- melt(corr_matrix)
ggplot(corr_melt, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile() +
  geom_text(aes(label = round(value, 2))) +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white", midpoint = 0) +
  labs(title = "Correlation Heatmap of Vehicle Variables", x = "", y = "") +
  theme_minimal()
ggsave("correlation_heatmap_vehicle.png", width = 7, height = 6)
cat("Interpretation: Variables most strongly correlated with Top_Speed influence overall
performance most.\n")
```

Output



Set 29: Student Academic Performance

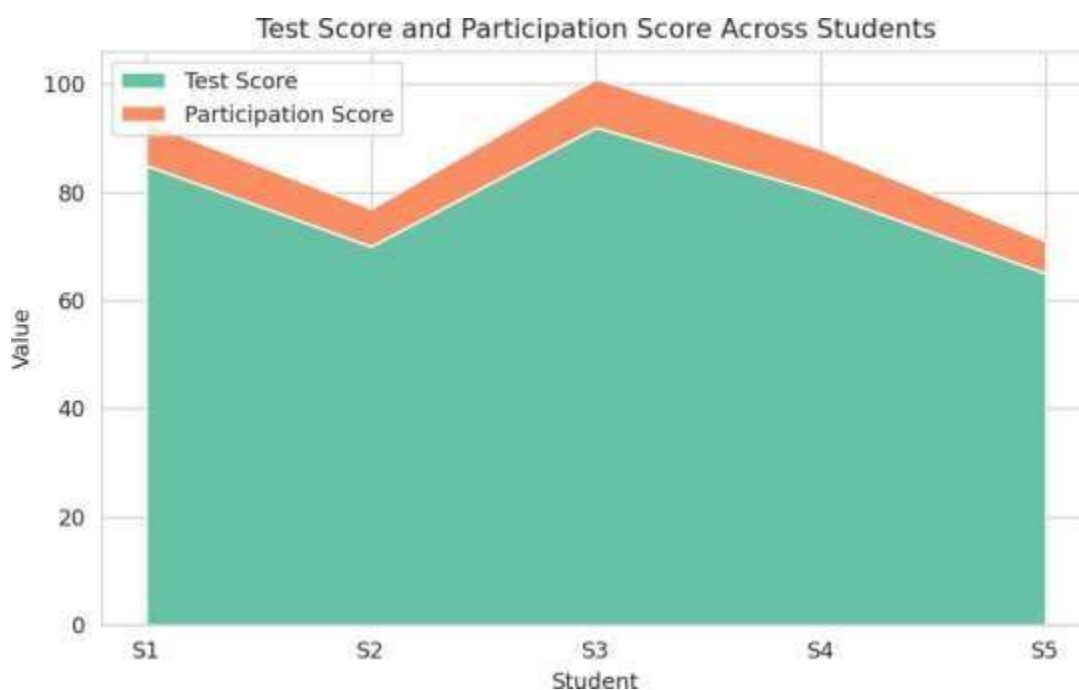
Question 1

Using R, create a stacked area chart showing *Test_Score* and *Participation_Score* across students. Analyze trends.

R Code

```
# 1. Stacked area chart - Test_Score and Participation_Score across students
df_long <- df %>%
  select(Student_ID, Test_Score, Participation_Score) %>%
  tidyr::pivot_longer(cols = c(Test_Score, Participation_Score), names_to = "Metric", values_to =
    "Value")
ggplot(df_long, aes(x = Student_ID, y = Value, fill = Metric, group = Metric)) +
  geom_area(position = "stack") +
  labs(title = "Test Score and Participation Score Across Students", x = "Student", y = "Value") +
  theme_minimal()
ggsave("stacked_area_scores.png", width = 7, height = 5)
```

Output



Question 2

Using R, generate a boxplot of *Study_Hours* grouped by *Attendance* quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

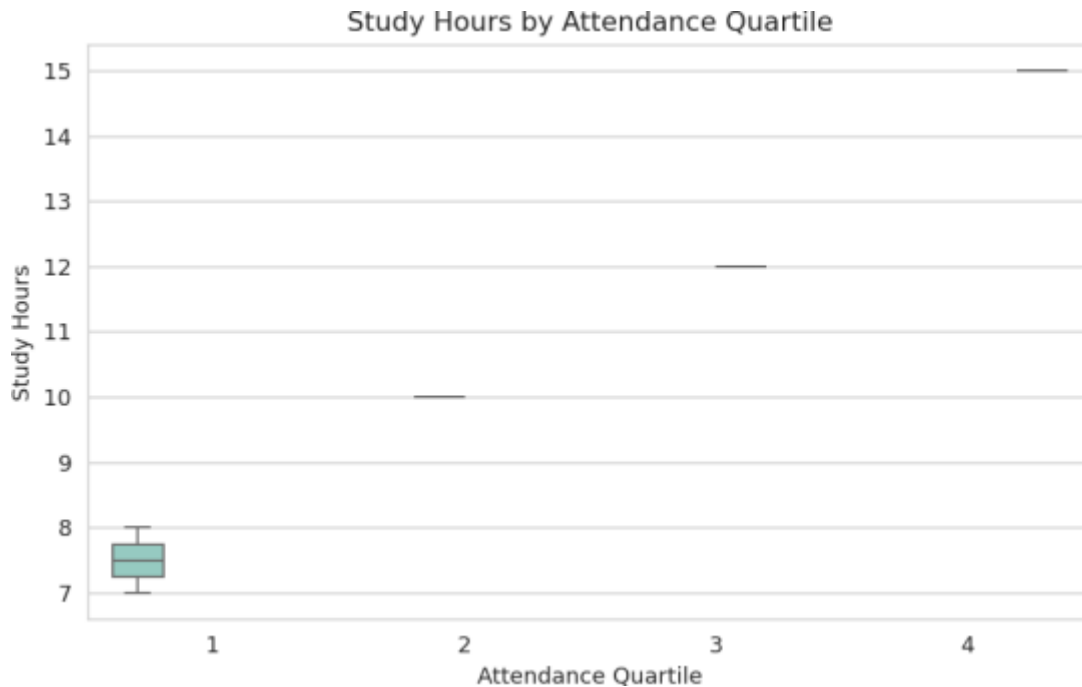
R Code

```
# 2. Boxplot of Study_Hours grouped by Attendance quartiles
df$AttendanceQuartile <- ntile(df$Attendance, 4)
```



```
ggplot(df, aes(x = factor(AttendanceQuartile), y = Study_Hours, fill =
factor(AttendanceQuartile))) +
geom_boxplot() +
labs(title = "Study Hours by Attendance Quartile", x = "Attendance Quartile", y = "Study Hours")
+
theme_minimal()
ggsave("boxplot_study_hours_by_attendance_quartile.png", width = 7, height = 5)
cat("Interpretation: Examine median and IQR of study hours per quartile and identify any
outliers.\n")
```

Output



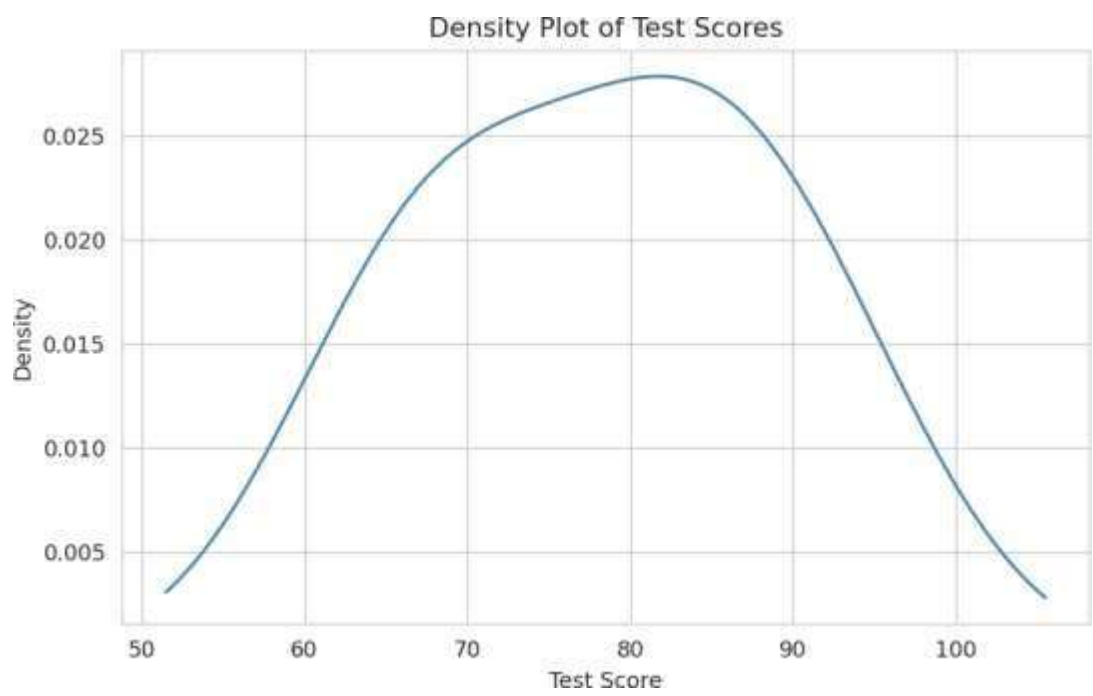
Question 3

Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

R Code

```
# 3. Density plot for Test_Score
ggplot(df, aes(x = Test_Score)) +
geom_density(fill = "lightblue", alpha = 0.6) +
labs(title = "Density Plot of Test Scores", x = "Test Score", y = "Density") +
theme_minimal()
ggsave("density_test_score.png", width = 7, height = 5)
```

Output



Set 30: Mobile App Usage

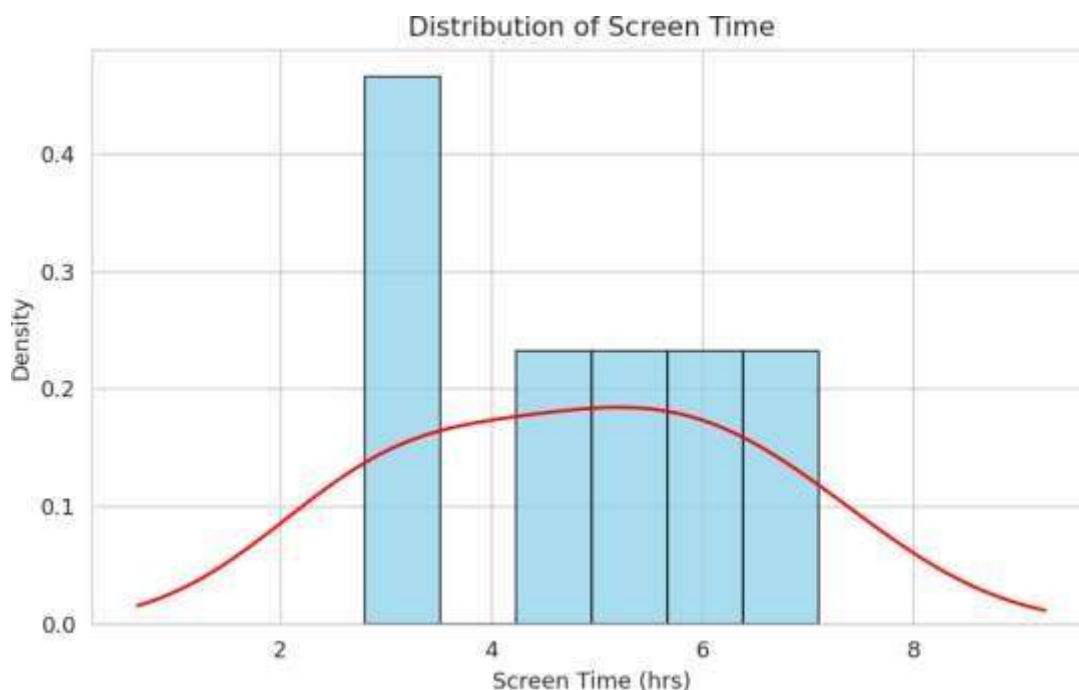
Question 1

Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

R Code

```
# 1. Histogram and density plot for Screen_Time
ggplot(df, aes(x = Screen_Time)) +
  geom_histogram(aes(y = after_stat(density)), binwidth = 1, fill = "skyblue", color = "black") +
  geom_density(color = "red", linewidth = 1) +
  labs(title = "Distribution of Screen Time", x = "Screen Time (hrs)", y = "Density") +
  theme_minimal()
ggsave("histogram_density_screen_time.png", width = 7, height = 5)
cat("Interpretation: Assess the shape/spread of screen-time distribution across users.\n")
```

Output



Question 2

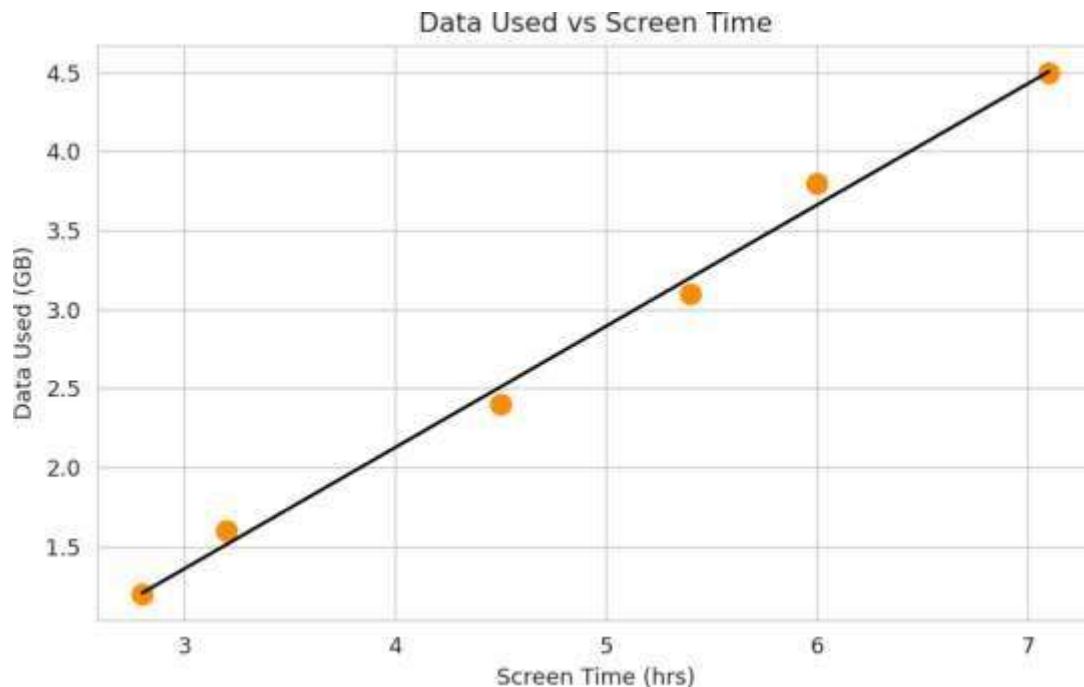
Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

R Code

```
# 2. Scatter plot - Data_Used vs Screen_Time with trend line, correlation
ggplot(df, aes(x = Screen_Time, y = Data_Used)) +
  geom_point(color = "darkorange", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "Data Used vs Screen Time", x = "Screen Time (hrs)", y = "Data Used (GB)") +
```

```
theme_minimal()
ggsave("scatter_data_used_vs_screen_time.png", width = 7, height = 5)
cat("Correlation between Data_Used and Screen_Time =", cor(df$Data_Used, df$Screen_Time), "\n")
```

Output



Question 3

Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

R Code

```
# 3. Average Satisfaction score for each Gender - bar chart with value labels
gender_avg <- df %>% group_by(Gender) %>% summarise(AvgSatisfaction = mean(Satisfaction))
ggplot(gender_avg, aes(x = Gender, y = AvgSatisfaction, fill = Gender)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = round(AvgSatisfaction, 2)), vjust = -0.5) +
  labs(title = "Average Satisfaction Score by Gender", x = "Gender", y = "Average Satisfaction") +
  theme_minimal()
ggsave("bar_chart_avg_satisfaction_by_gender.png", width = 7, height = 5)
cat("Interpretation: Compare average satisfaction levels between genders.\n")
```

Output

